Geophysics 3A — Week 10 Workshop

Geothermics: Finite Differences

April 9, 2026

Workshop Objectives

Students will:

- Derive finite difference equations from conservation laws.
- Implement steady and transient heat flow numerically.
- Understand numerical stability physically.

0:00–0:30 Heat Balance as Control Volumes

- Nodes as rock volumes.
- Fluxes into and out of nodes.
- Role of k, A, and κ.

0:30–1:15 Building a 1-D FD Model

- Discretizing space and time.
- Boundary and initial conditions.
- Steady-state vs transient solutions.

1:15–1:25 Break

1:25–2:00 When Analytic Solutions Still Work

- Smooth vs discontinuous properties.
- Time-varying boundary conditions.
- Geological messiness as motivation for numerics.

2:00–2:40 Numerical Stability and Information Transfer

Concept Stability limits impose a “speed limit” on thermal information.

$$\Delta t < \frac{\Delta x^2}{2\kappa}$$

- Large Δt: information skips nodes.
- Small Δx: CFL-like constraint.
- Analogy to finite signal propagation speed.

2:40–3:00 Wrap-Up

- Compare analytic and numeric solutions.
- Connect FD to earlier potential-field problems.
- Preview applications beyond heat flow.